

H&M 推荐系统本地验证与新特征消融实验报告

1. 实验背景

本报告基于期末大作业 Kaggle H&M 推荐系统项目完成两个小实验：

1. 防穿越的本地验证：使用时间顺序切分验证集，保证验证集时间晚于训练集时间，并记录最简单 Baseline 模型的 Local CV 与 Kaggle Public LB 分数。
2. 新特征消融实验：选取最近新增的 3 个 LightGBM 特征，遵循“一次只加一个变量”的原则，分别加入模型并记录 Local CV MAP@12 变化。

评价指标统一使用 Kaggle H&M 赛题指标 MAP@12 。

2. 防穿越的本地验证

2.1 验证集切分方式

本项目使用**按时间顺序切分**的本地验证方式，而不是随机打乱数据。代码中的验证逻辑为：

训练集：只使用验证窗口开始之前的历史交易

验证集：训练截止日之后未来 7 天的真实购买记录

例如最后一折验证为：

```
train_end    = 2020-09-15
valid_start  = 2020-09-16
valid_end    = 2020-09-22
```

这说明模型在训练时只能看到 2020-09-15 及以前的数据，再预测 2020-09-16 到 2020-09-22 的用户购买行为，符合“用过去预测未来”的推荐系统场景。

2.2 Baseline 本地验证结果

最简单 Baseline 使用 Final_Project.py 中的规则融合推荐器，并通过 Final_Project_Eval.py 进行权重调参和 5 折时间窗口验证。最终选择的规则权重为：

```
local_u40_l7_c5_d5_a5_t5.5_g1.2
user=40.0, long=7.0, channel=5.0, department=5.0, age=5.0, trend=5.5, global=1.2
```

5 折本地验证结果如下：

Fold	训练截止日	验证窗口	验证用户数	MAP@12
1	2020-08-18	2020-08-19 ~ 2020-08-25	72035	0.019575
2	2020-08-25	2020-08-26 ~ 2020-09-01	80253	0.018287
3	2020-09-01	2020-09-02 ~ 2020-09-08	75822	0.020984
4	2020-09-08	2020-09-09 ~ 2020-09-15	72019	0.022782
5	2020-09-15	2020-09-16 ~ 2020-09-22	68984	0.022402

Baseline 的 Local CV 与线上分数如下：

模型	本地验证方式	Local CV MAP@12	Public LB	Private LB
规则融合 Baseline	5 折时间顺序验证， 每折验证未来 7 天	0.020806	0.01828	0.01841

```
selected ranker: local_u40_l7_c5_d5_a5_t5.5_g1.2 (user=40.0, long=7.0, channel=5.0, department
=5.0, age=5.0, trend=5.5, global=1.2)
offline evaluation ranker: local_u40_l7_c5_d5_a5_t5.5_g1.2 (user=40.0, long=7.0, channel=5.0,
department=5.0, age=5.0, trend=5.5, global=1.2) (all_customers)
[Fold 1] train_end=2020-08-18 valid=2020-08-19~2020-08-25 customers=72035 MAP@12=0.019575
[Fold 2] train_end=2020-08-25 valid=2020-08-26~2020-09-01 customers=80253 MAP@12=0.018287
[Fold 3] train_end=2020-09-01 valid=2020-09-02~2020-09-08 customers=75822 MAP@12=0.020984
[Fold 4] train_end=2020-09-08 valid=2020-09-09~2020-09-15 customers=72019 MAP@12=0.022782
[Fold 5] train_end=2020-09-15 valid=2020-09-16~2020-09-22 customers=68984 MAP@12=0.022402
ranker tuning metrics saved: /kaggle/working/outputs/ranker_tuning_metrics.csv
offline fold metrics saved: /kaggle/working/outputs/offline_fold_metrics.csv
selected ranker saved: /kaggle/working/outputs/selected_ranker.csv
```



2.3 为什么不能使用随机打乱交叉验证

推荐系统预测的是用户未来的购买行为，用户兴趣、商品热度、季节趋势和新品生命周期都强烈依赖时间。如果把交易记录随机打乱做交叉验证，训练集中可能会混入验证时间之后的购买记录，模型等于提前看到了未来的热门商品、用户偏好变化和商品趋势，这会造成严重的数据穿越。

随机打乱交叉验证得到的本地分数通常会虚高，因为它不再模拟真实线上场景。线上提交时模型只能使用测试期之前的数据预测未来购买，因此本地验证也必须按时间顺序切分，保证训练集时间早于验证集时间。

3. 新特征单变量消融实验

3.1 实验设计

本实验选取最近新增的 3 个 LightGBM 特征：

特征	含义
item_pop_ratio_7d_30d	商品近 7 天销量与近 30 天销量的比例，用于刻画短期趋势增强
item_pop_ratio_30d_all	商品近 30 天销量与全历史销量的比例，用于刻画近期热度占比
price_diff_user_item	用户历史平均购买价格与商品平均价格的差值，用于刻画价格匹配程度

实验遵循“一次只加一个变量”的原则：

1. 先跑基础模型 base ，不包含以上 3 个新特征。
2. 在基础模型上只加入 item_pop_ratio_7d_30d 。
3. 在基础模型上只加入 item_pop_ratio_30d_all 。
4. 在基础模型上只加入 price_diff_user_item 。

实验中固定候选召回和 LightGBM 参数，只改变特征开关：

```
LGBM_FEATURE_EXPERIMENT=base
LGBM_FEATURE_EXPERIMENT=add_item_pop_ratio_7d_30d
LGBM_FEATURE_EXPERIMENT=add_item_pop_ratio_30d_all
LGBM_FEATURE_EXPERIMENT=add_price_diff_user_item
```

本地验证仍然采用时间顺序验证，避免未来数据穿越。

3.2 消融结果

4 次实验得到的 Local CV MAP@12 分数为：

base	= 0.030038
+ item_pop_ratio_7d_30d	= 0.030059
+ item_pop_ratio_30d_all	= 0.030730
+ price_diff_user_item	= 0.029983

以 base = 0.030038 为基础分数，计算每个新特征的边际变化：

特征名字	加入后的 Local CV MAP@12	分数变化量	决策
item_pop_ratio_7d_30d	0.030059	+0.000021	暂时保留，但贡献很小
item_pop_ratio_30d_all	0.030730	+0.000692	保留
price_diff_user_item	0.029983	-0.000055	删除 / 暂不加入最终模型

```
/kaggle/working
env: LGBM_MAX_CANDIDATES_PER_CUSTOMER=60
env: LGBM_BASELINE_RECALL_TOP=60
env: LGBM_ATTRIBUTE_RECALL_TOP=0
env: LGBM_RECENT_GLOBAL_RECALL_TOP=0
env: LGBM_ENABLE_COLOUR_RECALL=0
env: LGBM_ENABLE_SECTION_RECALL=0
env: LGBM_DROP_NOISY_FEATURES=0
env: LGBM_TRAIN_CUSTOMER_CAP=20000
env: LGBM_VALIDATION_CUSTOMER_CAP=20000
env: LGBM_N_ESTIMATORS=120
env: LGBM_FEATURE_EXPERIMENT=base
LGBM params: train_cap=20000, valid_cap=20000, max_candidates=60, baseline_recall_top=60, attribute_top=0, recent_global_top=0, n_estimators=120, colour_recall=False, section_recall=False, drop_noisy_features=False, feature_experiment=base
model feature count: 34 / raw feature count: 37
ranker cache not found, use fallback: hybrid_base (user=30.0, long=10.0, channel=6.0, department=6.0, age=5.0, trend=4.0, global=1.5)
candidate ranker: hybrid_base (user=30.0, long=10.0, channel=6.0, department=6.0, age=5.0, trend=4.0, global=1.5)
window dataset: history_end=2020-09-08 labels=2020-09-09~2020-09-15 customers=20000 rows=1258488
[LightGBM] [Info] Number of positive: 63563, number of negative: 1194925
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of testing was 0.087036 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4055
[LightGBM] [Info] Number of data points in the train set: 1258488, number of used features: 29
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.050507 -> initscore=-2.933807
[LightGBM] [Info] Start training from score -2.933807
LightGBM trained: rows=1258488 positives=63563
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
LGBM validation: history_end=2020-09-15 labels=2020-09-16~2020-09-22 customers=20000 MAP@12=0.030038
LGBM validation metrics saved: /kaggle/working/outputs/lgbm_validation_metrics.csv
```

```
/kaggle/working
env: LGBM_MAX_CANDIDATES_PER_CUSTOMER=60
env: LGBM_BASELINE_RECALL_TOP=60
env: LGBM_ATTRIBUTE_RECALL_TOP=0
env: LGBM_RECENT_GLOBAL_RECALL_TOP=0
env: LGBM_ENABLE_COLOUR_RECALL=0
env: LGBM_ENABLE_SECTION_RECALL=0
env: LGBM_DROP_NOISY_FEATURES=0
env: LGBM_TRAIN_CUSTOMER_CAP=20000
env: LGBM_VALIDATION_CUSTOMER_CAP=20000
env: LGBM_N_ESTIMATORS=120
env: LGBM_FEATURE_EXPERIMENT=add_item_pop_ratio_7d_30d
LGBM params: train_cap=20000, valid_cap=20000, max_candidates=60, baseline_recall_top=60, attribute_top=0, recent_global_top=0, n_estimators=120, colour_recall=False, section_recall=False, drop_noisy_features=False, feature_experiment=add_item_pop_ratio_7d_30d
model feature count: 35 / raw feature count: 37
ranker cache not found, use fallback: hybrid_base (user=30.0, long=10.0, channel=6.0, department=6.0, age=5.0, trend=4.0, global=1.5)
candidate ranker: hybrid_base (user=30.0, long=10.0, channel=6.0, department=6.0, age=5.0, trend=4.0, global=1.5)
window dataset: history_end=2020-09-08 labels=2020-09-09~2020-09-15 customers=20000 rows=1258483
[LightGBM] [Info] Number of positive: 63563, number of negative: 1194920
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of testing was 0.091007 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4318
[LightGBM] [Info] Number of data points in the train set: 1258483, number of used features: 30
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.050508 -> initscore=-2.933803
[LightGBM] [Info] Start training from score -2.933803
LightGBM trained: rows=1258483 positives=63563
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
LGBM validation: history_end=2020-09-15 labels=2020-09-16~2020-09-22 customers=20000 MAP@12=0.030059
LGBM validation metrics saved: /kaggle/working/outputs/lgbm_validation_metrics.csv
```

```
/kaggle/working
env: LGBM_MAX_CANDIDATES_PER_CUSTOMER=60
env: LGBM_BASELINE_RECALL_TOP=60
env: LGBM_ATTRIBUTE_RECALL_TOP=0
env: LGBM_RECENT_GLOBAL_RECALL_TOP=0
env: LGBM_ENABLE_COLOUR_RECALL=0
env: LGBM_ENABLE_SECTION_RECALL=0
env: LGBM_DROP_NOISY_FEATURES=0
env: LGBM_TRAIN_CUSTOMER_CAP=20000
env: LGBM_VALIDATION_CUSTOMER_CAP=20000
env: LGBM_N_ESTIMATORS=120
env: LGBM_FEATURE_EXPERIMENT=add_item_pop_ratio_30d_all
LGBM params: train_cap=20000, valid_cap=20000, max_candidates=60, baseline_recall_top=60, attribute_top=0, recent_global_top=0, n_estimators=120, colour_recall=False, section_recall=False, drop_noisy_features=False, feature_experiment=add_item_pop_ratio_30d_all
model feature count: 35 / raw feature count: 37
ranker cache not found, use fallback: hybrid_base (user=30.0, long=10.0, channel=6.0, department=6.0, age=5.0, trend=4.0, global=1.5)
candidate ranker: hybrid_base (user=30.0, long=10.0, channel=6.0, department=6.0, age=5.0, trend=4.0, global=1.5)
window dataset: history_end=2020-09-08 labels=2020-09-09~2020-09-15 customers=20000 rows=1258491
[LightGBM] [Info] Number of positive: 63563, number of negative: 1194928
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of testing was 0.073837 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4308
[LightGBM] [Info] Number of data points in the train set: 1258491, number of used features: 30
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.050507 -> initscore=-2.933810
[LightGBM] [Info] Start training from score -2.933810
LightGBM trained: rows=1258491 positives=63563
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
LGBM validation: history_end=2020-09-15 labels=2020-09-16~2020-09-22 customers=20000 MAP@12=0.030730
LGBM validation metrics saved: /kaggle/working/outputs/lgbm_validation_metrics.csv
```

+ Code

+ Markdown

```
/kaggle/working
env: LGBM_MAX_CANDIDATES_PER_CUSTOMER=60
env: LGBM_BASELINE_RECALL_TOP=60
env: LGBM_ATTRIBUTE_RECALL_TOP=0
env: LGBM_RECENT_GLOBAL_RECALL_TOP=0
env: LGBM_ENABLE_COLOUR_RECALL=0
env: LGBM_ENABLE_SECTION_RECALL=0
env: LGBM_DROP_NOISY_FEATURES=0
env: LGBM_TRAIN_CUSTOMER_CAP=20000
env: LGBM_VALIDATION_CUSTOMER_CAP=20000
env: LGBM_N_ESTIMATORS=120
env: LGBM_FEATURE_EXPERIMENT=add_price_diff_user_item
LGBM params: train_cap=20000, valid_cap=20000, max_candidates=60, baseline_recall_top=60, attribute_top=0, recent_global_top=0, n_estimators=120, colour_recall=False, section_recall=False, drop_noisy_features=False, feature_experiment=add_price_diff_user_item
model feature count: 35 / raw feature count: 37
ranker cache not found, use fallback: hybrid_base (user=30.0, long=10.0, channel=6.0, department=6.0, age=5.0, trend=4.0, global=1.5)
candidate ranker: hybrid_base (user=30.0, long=10.0, channel=6.0, department=6.0, age=5.0, trend=4.0, global=1.5)
window dataset: history_end=2020-09-08 labels=2020-09-09~2020-09-15 customers=20000 rows=1258502
[LightGBM] [Info] Number of positive: 63563, number of negative: 1194939
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of testing was 0.076996 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 4310
[LightGBM] [Info] Number of data points in the train set: 1258502, number of used features: 30
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.050507 -> initscore=-2.933819
[LightGBM] [Info] Start training from score -2.933819
LightGBM trained: rows=1258502 positives=63563
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
LGBM validation: history_end=2020-09-15 labels=2020-09-16~2020-09-22 customers=20000 MAP@12=0.029983
LGBM validation metrics saved: /kaggle/working/outputs/lgbm_validation_metrics.csv
```

3.3 消融结论

从实验结果看，`item_pop_ratio_30d_all` 带来了最明显的提升，Local CV MAP@12 从 `0.030038` 提升到 `0.030730`，变化量为 `+0.000692`，说明商品近期热度占全历史热度的比例对推荐排序有明显帮助，应当保留。

`item_pop_ratio_7d_30d` 的分数只提升了 `+0.000021`，贡献非常小，但方向为正，可以暂时保留，后续需要结合更多时间窗口继续验证其稳定性。`price_diff_user_item` 加入后分数下降 `-0.000055`，说明用户均价与商品均价差距在当前验证设置下没有带来收益，可能引入噪声，因此暂时不建议放入最终模型。

4. 最终结论

本次实验首先验证了项目中的本地验证流程是防穿越的：训练集始终早于验证集，验证窗口模拟未来 7 天购买行为，符合 H&M 推荐系统“用过去预测未来”的业务场景。规则融合 Baseline 的本地 5 折时间验证 MAP@12 为 `0.020806`，线上 Public LB 为 `0.01828`。

新特征消融实验表明，3 个新增特征中最有价值的是 `item_pop_ratio_30d_all`，它显著提升了本地验证 MAP@12；`item_pop_ratio_7d_30d` 只有极小正贡献，可以暂时保留；`price_diff_user_item` 在本地验证中造成分数下降，建议删除或暂不用于最终模型。

因此，后续模型优化应优先保留与商品近期热度相关的特征，谨慎使用价格匹配类特征，并继续坚持时间顺序验证，避免随机打乱带来的数据穿越和分数虚高。